Year 11 Foundation

REVISION PACK

Progress Check 2

ANSWERS

Units in this pack

- Graphs 5
- Handling Data 4

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Graphs 5 · Simultaneous equations (Lesson 3.1)

Q	Working	Answer
adding the equations		
1.1	$x + y = 12$ $x - y = 4$ $\text{add}: \ 2x = 16$ $x = 8$ $1 \times 8 + y = 12$ $y = 4$	$x = 8, y = 4$
1.2	$x + y = 11$ $x - y = 7$ $\text{add}: \ 2x = 18$ $x = 9$ $1 \times 9 + y = 11$ $y = 2$	$x = 9, y = 2$
1.3	$x + y = 12$ $x - y = 2$ $\text{add}: \ 2x = 14$ $x = 7$ $1 \times 7 + y = 12$ $y = 5$	$x = 7, y = 5$
1.4	$x + y = 13$ $x - y = 7$ $\text{add}: \ 2x = 20$ $x = 10$ $1 \times 10 + y = 13$ $y = 3$	$x = 10, y = 3$

1.5	$x + y = 7$ $x - y = 5$ $\text{add}: \ 2x = 12$ $x = 6$ $1 \times 6 + y = 7$ $y = 1$	$x = 6, y = 1$
1.6	$x + y = 15$ $x - y = 7$ $\text{add}: \ 2x = 22$ $x = 11$ $1 \times 11 + y = 15$ $y = 4$	$x = 11, y = 4$
subtracting the equations		
1.7	$2x + y = 13$ $x + y = 8$ $\text{subtract}: \ x = 5$ $2 \times 5 + y = 13$ $y = 3$	$x = 5, y = 3$
1.8	$4x + y = 14$ $x + y = 5$ $\text{subtract}: \ 3x = 9$ $x = 3$ $4 \times 3 + y = 14$ $y = 2$	$x = 3, y = 2$
1.9	$3x + y = 19$ $x + y = 7$ $\text{subtract}: \ 2x = 12$ $x = 6$ $3 \times 6 + y = 19$ $y = 1$	$x = 6, y = 1$

1.10	$5x + y = 17$ $x + y = 9$ subtract : $4x = 8$ $x = 2$ $5 \times 2 + y = 17$ $y = 7$	$x = 2, y = 7$
1.11	$2x + y = 20$ $x + y = 12$ subtract : $x = 8$ $2 \times 8 + y = 20$ $y = 4$	$x = 8, y = 4$
1.12	$3x + y = 21$ $x + y = 13$ subtract : $2x = 8$ $x = 4$ $3 \times 4 + y = 21$ $y = 9$	$x = 4, y = 9$
solving using graphs		
1.13	where the lines cross $(3, 5)$	$x = 3, y = 5$
1.14	where the lines cross $(2, 3)$	$x = 2, y = 3$
1.15	where the lines cross $(4, 3)$	$x = 4, y = 3$
1.16	where the lines cross $(2, 4)$	$x = 2, y = 4$
adding with a number in front of x		

1.17	$2x + y = 12$ $x - y = 3$ $\text{add}: \ 3x = 15$ $x = 5$ $2 \times 5 + y = 12$ $y = 2$	$x = 5, y = 2$
1.18	$4x + y = 13$ $x - y = 2$ $\text{add}: \ 5x = 15$ $x = 3$ $4 \times 3 + y = 13$ $y = 1$	$x = 3, y = 1$
1.19	$3x + y = 11$ $x - y = -3$ $\text{add}: \ 4x = 8$ $x = 2$ $3 \times 2 + y = 11$ $y = 5$	$x = 2, y = 5$
1.20	$5x + y = 11$ $x - y = 1$ $\text{add}: \ 6x = 12$ $x = 2$ $5 \times 2 + y = 11$ $y = 1$	$x = 2, y = 1$
1.21	$2x + y = 15$ $x - y = 3$ $\text{add}: \ 3x = 18$ $x = 6$ $2 \times 6 + y = 15$ $y = 3$	$x = 6, y = 3$

1.22	$4x + y = 10$ $x - y = -5$ $\text{add}: \ 5x = 5$ $x = 1$ $4 \times 1 + y = 10$ $y = 6$	$x = 1, y = 6$

Handling Data 4 · Collecting and presenting data (Lessons 4.1, 4.2, 4.3, 4.4)

Q	Working	Answer
frequency tables		
2.1	$(a)\ 0:3,\ 1:8,\ 2:6,\ 3:2,\ 4:1$ $(c)\ 4 - 0 = 4$ $(d)\ 0 \times 3 + 1 \times 8 + 2 \times 6 + 3 \times 2 + 4 \times 1 = 30$ $30 \div 20 = 1.5$	**(a) frequencies 3, 8, 6, 2, 1** **(b) 1** **(c) 4** **(d) 1.5**
2.2	$(a)\ 0:2,\ 1:4,\ 2:8,\ 3:4,\ 4:2$ $(c)\ 4 - 0 = 4$ $(d)\ 0 \times 2 + 1 \times 4 + 2 \times 8 + 3 \times 4 + 4 \times 2 = 40$ $40 \div 20 = 2$	**(a) frequencies 2, 4, 8, 4, 2** **(b) 2** **(c) 4** **(d) 2**
2.3	$(a)\ 0:4,\ 1:9,\ 2:4,\ 3:2,\ 4:1$ $(c)\ 4 - 0 = 4$ $(d)\ 0 \times 4 + 1 \times 9 + 2 \times 4 + 3 \times 2 + 4 \times 1 = 27$ $27 \div 20 = 1.35$	**(a) frequencies 4, 9, 4, 2, 1** **(b) 1** **(c) 4** **(d) 1.35**
2.4	$(a)\ 0:2,\ 1:5,\ 2:7,\ 3:4,\ 4:2$ $(c)\ 4 - 0 = 4$ $(d)\ 0 \times 2 + 1 \times 5 + 2 \times 7 + 3 \times 4 + 4 \times 2 = 39$ $39 \div 20 = 1.95$	**(a) frequencies 2, 5, 7, 4, 2** **(b) 2** **(c) 4** **(d) 1.95**
2.5	$(a)\ 0:3,\ 1:9,\ 2:5,\ 3:2,\ 4:1$ $(c)\ 4 - 0 = 4$ $(d)\ 0 \times 3 + 1 \times 9 + 2 \times 5 + 3 \times 2 + 4 \times 1 = 29$ $29 \div 20 = 1.45$	**(a) frequencies 3, 9, 5, 2, 1** **(b) 1** **(c) 4** **(d) 1.45**

2.6	$(a)\ 0:5,\ 1:9,\ 2:4,\ 3:1,\ 4:1$ $(c)\ 4-0=4$ $(d)\ 0\times5+1\times9+2\times4+3\times1+4\times1=24$ $24\div20=1.2$	**(a) frequencies 5, 9, 4, 1, 1** **(b) 1** **(c) 4** **(d) 1.2**
pie charts from a table		
2.7	$360\div30=12$ $7\times12=84$	**84°**
2.8	$360\div24=15$ $10\times15=150$	**150°**
2.9	$360\div40=9$ $14\times9=126$	**126°**
2.10	$360\div18=20$ $5\times20=100$	**100°**
2.11	$360\div45=8$ $12\times8=96$	**96°**
2.12	$360\div60=6$ $22\times6=132$	**132°**
the median of six numbers		
2.13	$2,4,5,7,8,10$ $(5+7)\div2=6$	**6**
2.14	$3,4,6,9,12,15$ $(6+9)\div2=7.5$	**7.5**
2.15	$1,3,5,6,7,9$ $(5+6)\div2=5.5$	**5.5**
2.16	$6,8,10,11,12,14$ $(10+11)\div2=10.5$	**10.5**
2.17	$1,2,4,5,9,9$ $(4+5)\div2=4.5$	**4.5**

2.18	$11, 13, 15, 16, 18, 20$ $(15 + 16) \div 2 = 15.5$	**15.5**
misleading charts		
2.19		**The vertical axis does not start at 0 (it starts at 50), so the differences look bigger than they are.**
2.20		**The vertical axis does not start at 0 (it starts at 80), so the differences look bigger than they are.**
2.21		**The vertical axis does not start at 0 (it starts at 120), so the differences look bigger than they are.**
2.22		**The vertical axis does not start at 0 (it starts at 40), so the differences look bigger than they are.**
2.23		**The vertical axis does not start at 0 (it starts at 60), so the differences look bigger than they are.**
2.24		**The vertical axis does not start at 0 (it starts at 30), so the differences look bigger than they are.**
questionnaires		
2.25		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 6; no box for 0**
2.26		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 7; no box for 0**
2.27		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 6 KD; no box for 0**
2.28		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 6; no box for 0**

2.29		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 30 KD; no box for 0**
2.30		**Any two of: no time period (for example per day or per week); the boxes overlap; no box for more than 4; no box for 0**
pie charts		
2.31	$360 \div 30 = 12$ $11 \times 12 = 132$	**132°**
2.32	$360 \div 45 = 8$ $16 \times 8 = 128$	**128°**
2.33	$360 \div 60 = 6$ $25 \times 6 = 150$	**150°**
2.34	$360 \div 40 = 9$ $9 \times 9 = 81$	**81°**
2.35	$360 \div 72 = 5$ $20 \times 5 = 100$	**100°**
2.36	$360 \div 20 = 18$ $7 \times 18 = 126$	**126°**

Handling Data 4 · Averages and range (Lesson 4.5)

Q	Working	Answer
mean and range		
3.1	**(a)** $7 + 3 + 9 + 5 + 6 = 30$ $30 \div 5 = 6$ **(b)** $9 - 3 = 6$	**(a) 6** **(b) 6**
3.2	**(a)** $12 + 8 + 15 + 10 + 5 = 50$ $50 \div 5 = 10$ **(b)** $15 - 5 = 10$	**(a) 10** **(b) 10**

3.3	**(a)** $2+9+4+7+3=25$ $25 \div 5 = 5$ **(b)** $9-2=7$	**(a) 5** **(b) 7**
3.4	**(a)** $6+11+8+10+5=40$ $40 \div 5 = 8$ **(b)** $11-5=6$	**(a) 8** **(b) 6**
3.5	**(a)** $14+9+12+7+8=50$ $50 \div 5 = 10$ **(b)** $14-7=7$	**(a) 10** **(b) 7**
3.6	**(a)** $3+10+5+8+4=30$ $30 \div 5 = 6$ **(b)** $10-3=7$	**(a) 6** **(b) 7**
mode and median		
3.7	**(a)** 4 is the most common **(b)** $4,4,4,5,6,8,9$ middle $= 5$	**(a) 4** **(b) 5**
3.8	**(a)** 3 is the most common **(b)** $3,3,3,5,6,7,9$ middle $= 5$	**(a) 3** **(b) 5**
3.9	**(a)** 8 is the most common **(b)** $1,2,4,5,8,8,8$ middle $= 5$	**(a) 8** **(b) 5**
3.10	**(a)** 7 is the most common **(b)** $7,7,7,9,10,11,12$ middle $= 9$	**(a) 7** **(b) 9**
3.11	**(a)** 2 is the most common **(b)** $2,2,2,3,5,6,9$ middle $= 3$	**(a) 2** **(b) 3**

3.12	**(a)**	15 is the most common	**(a) 15** **(b) 15**
	(b)	$11, 12, 14, 15, 15, 15, 18$ $\text{middle} = 15$	